

Security+ 2.4 · Other Malware Types

Mestre CyberSec · Lesson 2.4 — Other Malware Types (7:32) · [Assistir Vídeo no YouTube](#) · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: Keylogger: grava digitação — rouba senha em texto claro. Logic bomb: detonada por evento (data, demissão, tecla) — dorme até o gatilho. Rootkit: esconde no KERNEL — invisível pra antivírus comum (precisa de limpeza/reinstall). Trojan: finge ser legítimo. Backdoor: porta que o ata

💡 Key Exam Highlights & Critical Concepts

- Rootkit esconde no KERNEL do sistema: quase invisível (boot from known-good media pra detectar)
- Keylogger: grava tudo em texto claro — rouba senha no envio
- Logic bomb: detonada por EVENTO (data/demissão/action)

🎯 Exam Tips & Typical CompTIA Scenarios

- 'malware that hides itself within the OS kernel' = rootkit
- 'dormant until triggered by a specific event' = logic bomb

Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 2.4 — Other Malware Types)

Why is this advanced coverage required for SY0-701? Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• Keylogger (N/A)

Software or hardware that records every keystroke typed by a user, capturing credentials and sensitive data in plaintext before encryption.

 **Exam Trap:** Exam trigger: 'captures keystrokes in plaintext' or 'records what a user types'. Hardware keyloggers are physically inline between keyboard and motherboard; software keyloggers are often rootkit-based.

• Logic Bomb (N/A)

Malicious code that remains dormant until a specific trigger condition is met (date, time, event, user action, or termination), then executes its payload.

 **Exam Trap:** Exam trigger: 'dormant until triggered by a specific event' or 'executes when condition is met'. Distinct from time bombs (date-based) vs. event-based triggers like employee termination.

• Rootkit (N/A)

Malware that embeds itself deep within the operating system kernel or boot sector to maintain privileged, stealthy access and evade standard antivirus detection.

 **Exam Trap:** Exam trigger: 'malware that hides within the OS kernel' or 'invisible to standard AV'. Detection requires boot from known-good/clean media. Often necessitates full OS reinstallation.

• Trojan (Trojan Horse) (N/A)

Malware disguised as legitimate, useful software to trick users into installing it, then delivering a hidden malicious payload.

 **Exam Trap:** Exam trigger: 'disguised as legitimate software' or 'user must be tricked into running it'. Does NOT self-replicate (unlike viruses/worms). Classic example: fake antivirus or game cracks.

• Backdoor (N/A)

An undocumented or hidden access point in software or a system that bypasses normal authentication, allowing remote attacker access.

 **Exam Trap:** Exam trigger: 'unauthorized access point' or 'bypasses normal authentication'. Often left by developers OR installed by Trojans. Closely linked with RATs (Remote Access Trojans).

• Spyware (N/A)

Malware that covertly gathers information about a user's activities, browsing habits, credentials, or personal data without consent.

 **Exam Trap:** Exam trigger: 'covertly gathers user information' or 'monitors browsing habits'. Distinguished from keyloggers (broader data capture) and adware (specifically ad-driven).

• Adware (N/A)

Software that automatically displays or downloads unwanted advertisements, often bundled with free software and potentially collecting marketing data.

 **Exam Trap:** Exam trigger: 'displays unwanted advertisements' or 'bundled with free software'. Not always malicious but crosses into malware when it tracks behavior without consent (grayware).

• Ransomware (N/A)

Malware that encrypts user files or locks the system, demanding payment (ransom) typically in cryptocurrency for decryption keys.

⚠ Exam Trap: Exam trigger: 'encrypts files and demands payment' or 'crypto-malware'. Modern variants use double/triple extortion (encrypt + leak + DDoS). Backups are the primary defense.

✓ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

A financial analyst reports that credentials for an internal banking portal were stolen even though the portal uses HTTPS. Forensic review shows a small user-mode process on the analyst's workstation intercepting keyboard input before the browser encrypts it. Which malware type best matches this behavior?

- A) Keylogger
- B) Rootkit
- C) Logic bomb
- D) Adware

QUESTION 2 · EXAM SCENARIO

A former contractor leaves a script on a file server. The script checks every hour and, if the contractor's HR record is marked terminated, deletes project archives and disables backups. The code remains dormant for weeks until that condition occurs. Which malware type is this?

- A) Ransomware
- B) Rootkit
- C) Logic bomb
- D) Trojan horse

QUESTION 3 · EXAM SCENARIO

During incident response, analysts find that a compromised Linux server reports normal process lists and antivirus scans show no infection. However, kernel system call tables have been modified, and a malicious driver hides files and network connections. Which malware type is most likely present?

- A) Keylogger
- B) Rootkit
- C) Adware
- D) Logic bomb

QUESTION 4 · EXAM SCENARIO

An employee downloads what appears to be a free PDF-to-Word converter from a pop-up advertisement. The program converts PDFs as promised, but it also installs a remote access service that listens for commands from an external IP address. Which malware type best describes the converter?

- A) Keylogger
- B) Ransomware
- C) Logic bomb
- D) Trojan horse

QUESTION 5 · EXAM SCENARIO

After a workstation is cleaned of a previous malware infection, the security team observes recurring inbound connections to a high TCP port. Further analysis shows a hidden service that accepts commands using a hard-coded credential, allowing access without exploiting the original vulnerability again. What mechanism is this?

- A) Backdoor
- B) Rootkit
- C) Spyware
- D) Logic bomb

QUESTION 6 · EXAM SCENARIO

An endpoint security tool alerts on a process that periodically sends browsing history, clipboard contents, and application window titles to an external server. Users report no pop-up ads, and no files have been encrypted. Which malware category best matches this activity?

- A) Adware
- B) Spyware
- C) Ransomware
- D) Logic bomb

QUESTION 7 · EXAM SCENARIO

Employees who installed a free screensaver package report frequent pop-up advertisements, changed browser homepage, and new search toolbars. No credentials or documents appear to have been stolen. Which malware category is most likely involved?

- A) Spyware
- B) Keylogger
- C) Adware
- D) Rootkit

QUESTION 8 · EXAM SCENARIO

A hospital's shared drive files are replaced with encrypted versions that have a .locked extension. A text file on each affected system demands cryptocurrency payment in exchange for a decryption key. Which malware type is responsible?

- A) Rootkit
- B) Trojan horse
- C) Adware
- D) Ransomware

QUESTION 9 · EXAM SCENARIO

A compromised server is confirmed to have kernel-level malicious code that hides processes and modifies system calls. The operating system cannot be trusted, and the incident response team needs the most reliable way to remove the threat. What should the team do?

- A)** Run a full antivirus scan and delete detected files
- B)** Wipe the server and reinstall the OS from trusted media
- C)** Remove the malicious process from Task Manager
- D)** Change all local administrator passwords

QUESTION 10 · EXAM SCENARIO

A security analyst suspects a workstation is capturing user credentials before they are encrypted by the browser. The analyst has both host-based and network monitoring available. Which data source is most likely to reveal this activity?

- A)** Network packet capture filtering only HTTPS traffic
- B)** Perimeter firewall logs showing blocked inbound connections
- C)** Host-based API and keystroke monitoring on the workstation
- D)** DNS query logs for the banking portal domain

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option A — Keylogger

A keylogger captures keystrokes at the input layer before TLS/HTTPS encryption, so it can steal credentials in clear text. Rootkits hide at kernel level, logic bombs trigger on specific events, and adware displays unwanted advertising. HTTPS does not protect against local keystroke capture.

Question 2: Option C — Logic bomb

A logic bomb is dormant code that executes a malicious payload when a specific trigger occurs. Here the trigger is the HR termination status, after which it deletes archives and disables backups. Ransomware demands payment, rootkits hide in the kernel, and Trojans disguise themselves as legitimate software.

Question 3: Option B — Rootkit

A rootkit often modifies kernel components or system calls to conceal files, processes, and connections from normal tools and antivirus. Keyloggers capture input, adware displays ads, and logic bombs trigger on events. Kernel-level hiding points to a rootkit.

Question 4: Option D — Trojan horse

A Trojan horse appears to perform a legitimate function while secretly performing malicious actions. The converter works as advertised but also installs a remote access service. A keylogger captures keystrokes, ransomware encrypts for payment, and a logic bomb waits for a trigger.

Question 5: Option A — Backdoor

A backdoor is a hidden access mechanism that lets an attacker reconnect without re-exploiting the original vulnerability. It may use a listening service and hard-coded credentials. A rootkit conceals activity, spyware collects information, and a logic bomb triggers on an event.

Question 6: Option B — Spyware

Spyware covertly collects user activity and data, such as browsing history, clipboard contents, and window titles. Adware focuses on unwanted advertisements, ransomware encrypts files for payment, and logic bombs trigger a payload on an event. The stealthy data collection is spyware.

Question 7: Option C — Adware

Adware displays unwanted advertisements and often changes browser settings, such as the homepage and search toolbars. Spyware stealthily collects information, a keylogger captures keystrokes, and a rootkit hides at the kernel level. The ad-focused symptoms point to adware.

Question 8: Option D — Ransomware

Ransomware encrypts data and demands payment for a decryption key. A Trojan may deliver it, but the described payload is extortion through encryption. Rootkits hide activity, and adware displays advertisements.

Question 9: Option B — Wipe the server and reinstall the OS from trusted media

Kernel-level compromise makes the operating system untrustworthy, so the most reliable remediation is to wipe the server and reinstall the OS from trusted media. Antivirus may not detect kernel-hidden code, killing a process is insufficient, and changing passwords does not remove the malicious kernel components.

Question 10: Option C — Host-based API and keystroke monitoring on the workstation

If credentials are captured before browser encryption, the activity occurs locally on the workstation. Host-based API and keystroke monitoring can detect this. HTTPS packet capture would show encrypted traffic, firewall logs show blocked connections, and DNS logs show domain lookups but not keystrokes.